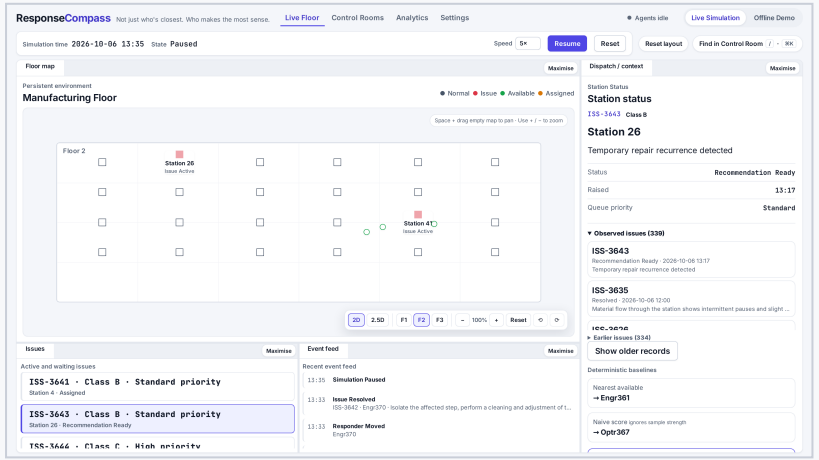
ResponseCompass

The right responder. The right fix.

Learned from every past repair.



Live Floor, simulated plant e8-aura-p1

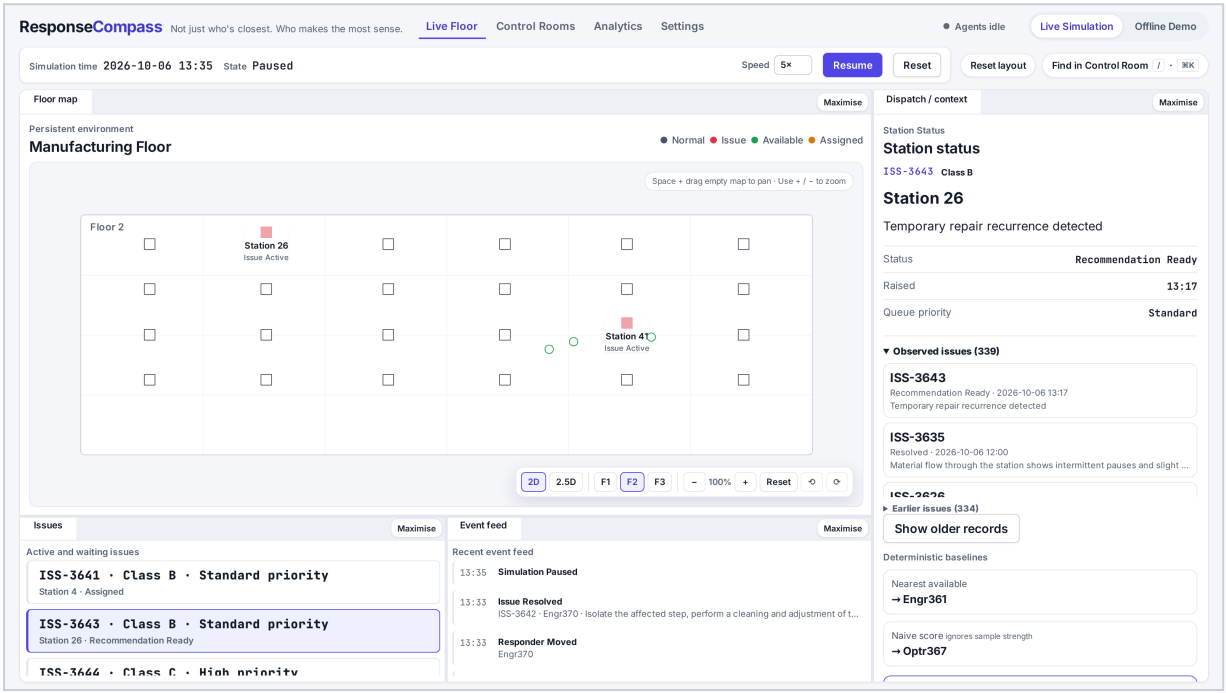
An AI dispatch agent for operations control rooms, built on AMD Token Factory.

The problem

- When equipment fails, **closest** or **free** may be an incomplete answer.
- The nearest technician may never have fixed this kind of fault.
- A poor match can mean downtime, a reopened issue or another trip.
- Who fixes what, and how, lives in people's heads and old tickets.

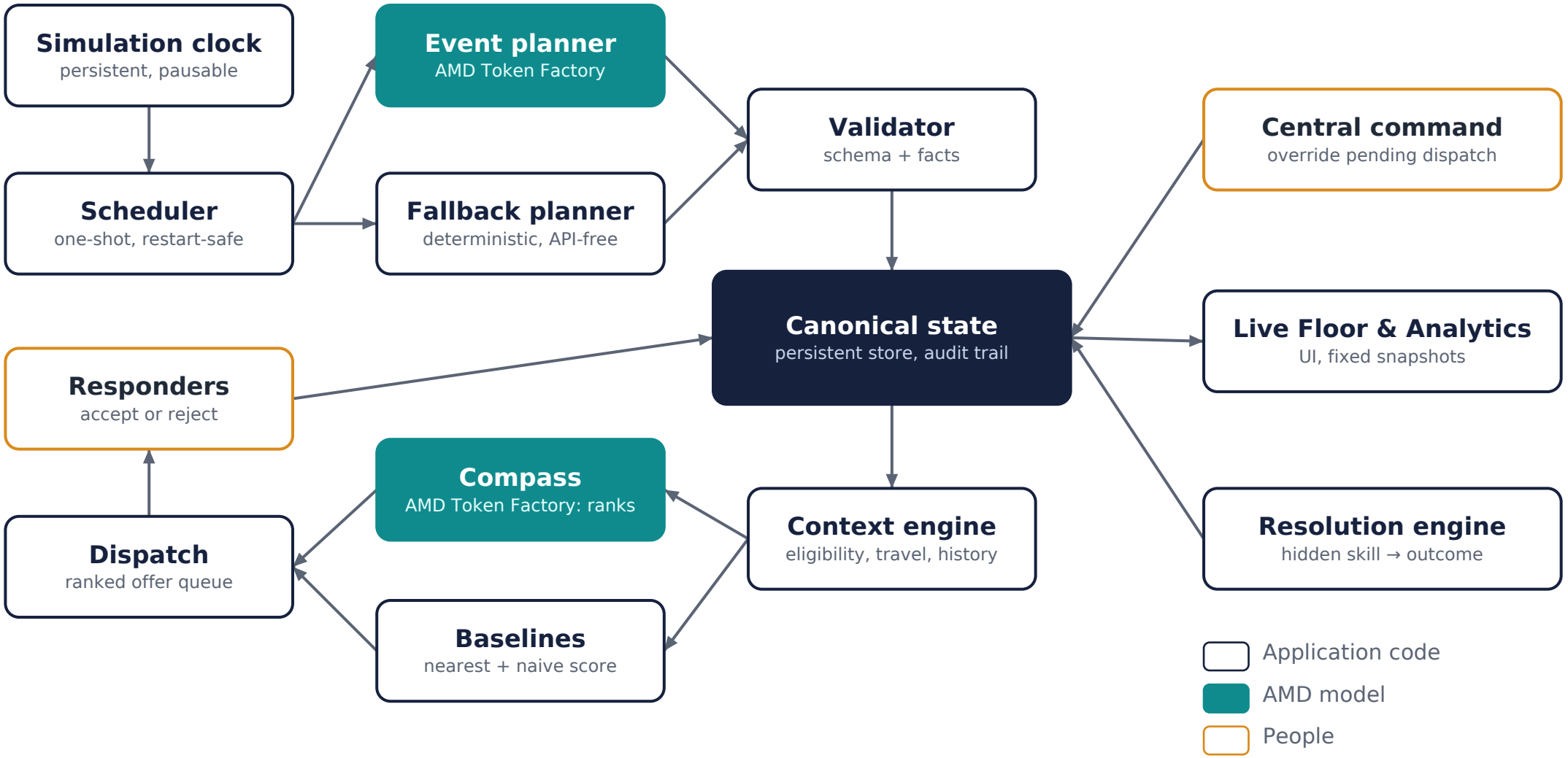
What ResponseCompass does

- For every new issue, Compass **ranks the eligible responders** and explains the tradeoffs.
- Offers go out automatically; responders accept or reject; complex work can get a second responder.
- Central command can override a pending dispatch; every step is audited.
- Each issue carries a **response action**; recommending fixes from past repairs is **planned**.



Code owns the facts. Models only rank.

Application code owns time, facts, eligibility, baselines and every mutation; models only propose.

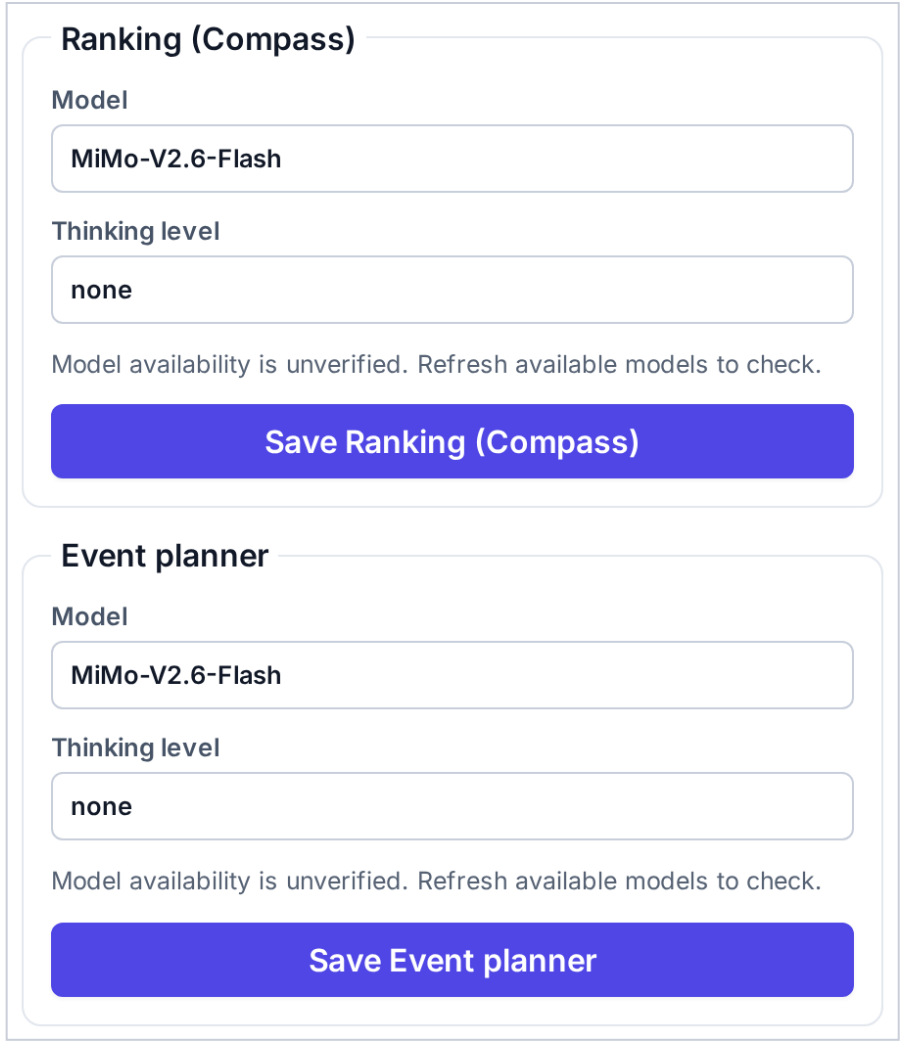


Honest by design

- Each responder's **true skill is hidden** simulation truth: no model, UI or analytics sees it.
- Outcomes are simulated from that hidden skill, independently of what the model claims.
- Two transparent baselines (nearest, naive score) run alongside; disagreements are flagged.
- Analytics show denominators, suppress thin samples and are not an employee leaderboard.
- Excluded from ranking: pay, seniority, demographics, fatigue, HR data.

Built on AMD Token Factory

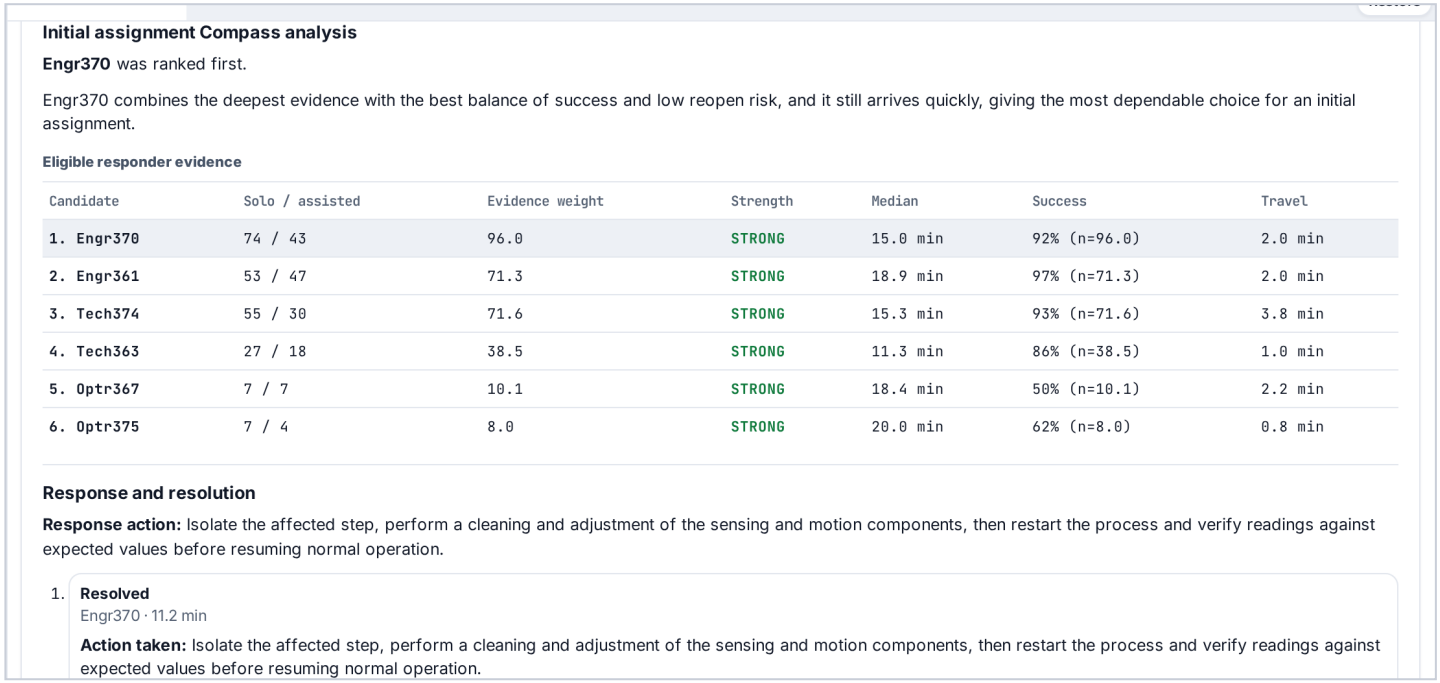
- OpenAI-compatible API, JSON output; our validators are the contract.
- Default model **MiMo-V2.6-Flash**; ranking and event planner can use different models.
- Local budget: 10 requests/min, 2 concurrent; bounded 429 retries; circuit breaker.
- Recorded MiMo check: **9/10** valid rankings, **10/10** valid event plans.
- Ranking deadline 90 s; model calls never hold the simulation clock.



A real MiMo ranking, end to end

ISS-3642, captured live on AMD Token Factory (MiMo-V2.6-Flash)

- Nearest responder: **Optr375**, 0.8 min away, thin history on this fault.
- Compass ranked **Engr370** first: deepest evidence, 92% success, 2 min away.
- Offer accepted; resolved first time in **11.2 min**.
- Every step, and the action taken, is in the issue's audit trail.



Initial assignment Compass analysis

Engr370 was ranked first.

Engr370 combines the deepest evidence with the best balance of success and low reopen risk, and it still arrives quickly, giving the most dependable choice for an initial assignment.

Eligible responder evidence

Candidate	Solo / assisted	Evidence weight	Strength	Median	Success	Travel
1. Engr370	74 / 43	96.0	STRONG	15.0 min	92% (n=96.0)	2.0 min
2. Engr361	53 / 47	71.3	STRONG	18.9 min	97% (n=71.3)	2.0 min
3. Tech374	55 / 30	71.6	STRONG	15.3 min	93% (n=71.6)	3.8 min
4. Tech363	27 / 18	38.5	STRONG	11.3 min	86% (n=38.5)	1.0 min
5. Optr367	7 / 7	10.1	STRONG	18.4 min	50% (n=10.1)	2.2 min
6. Optr375	7 / 4	8.0	STRONG	20.0 min	62% (n=8.0)	0.8 min

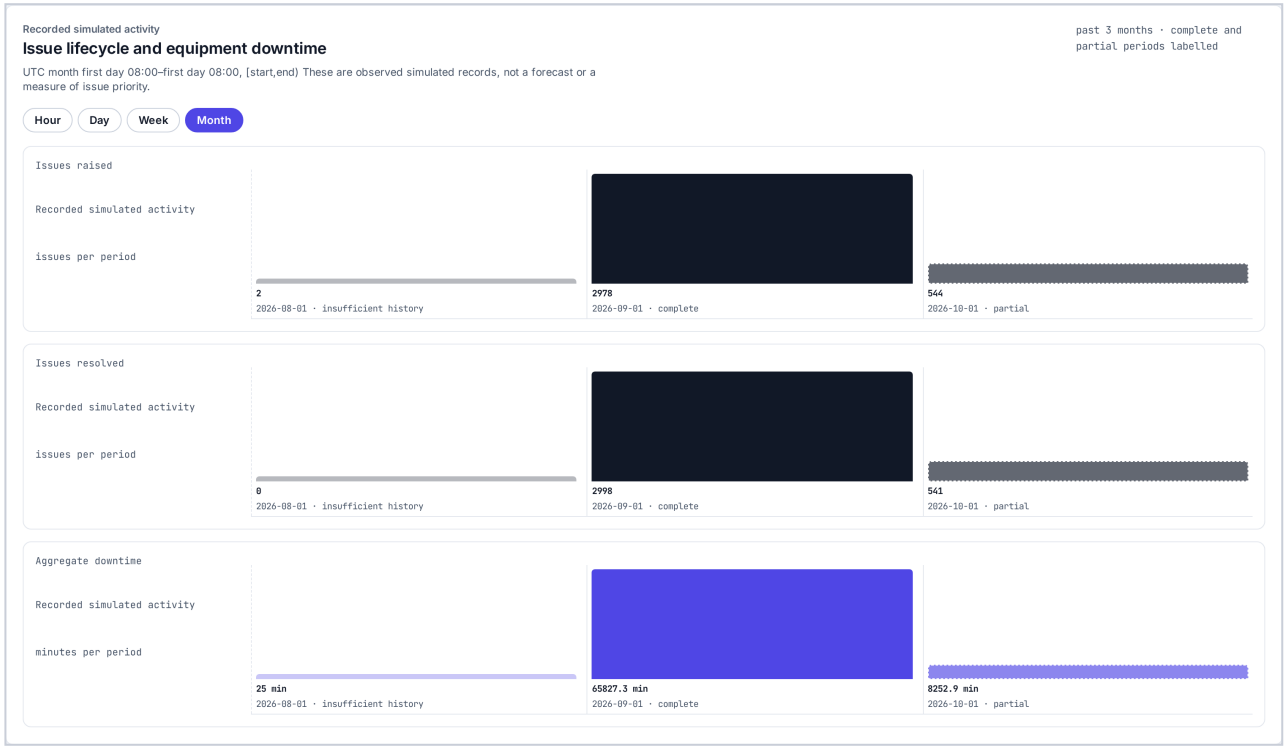
Response and resolution

Response action: Isolate the affected step, perform a cleaning and adjustment of the sensing and motion components, then restart the process and verify readings against expected values before resuming normal operation.

1. **Resolved**
   Engr370 · 11.2 min
   Action taken: Isolate the affected step, perform a cleaning and adjustment of the sensing and motion components, then restart the process and verify readings against expected values before resuming normal operation.

A month of operations

- e8-aura-p1: **3,531 issues**, 1 September to 6 October (simulated).
- September: **2,978 raised / 2,998 resolved** events.
- Day, week and month views; Compass-vs-baseline records.
- Seven-day detail retention: live state served in **56 ms** in a local check.



Engineering quality

- Live and Offline are one app; Offline makes no API calls and is fully reproducible.
- **277** hermetic unit and integration tests and **51** Chromium end-to-end tests.
- Deterministic seeded simulation; restart-safe scheduler; atomic, crash-safe persistence.
- Runs on Node.js 22 or Docker; no external services besides the model API.

What's next

- **Recommended fixes:** rank response actions retrieved from comparable past resolutions.
- Learn from outcomes: feed accepted and successful fixes back into the evidence.
- Connect to real maintenance and ticketing systems in place of the simulated floor.
- Long-run storage: compact the remaining growing history for year-long runs.

The right responder. The right fix.